

Advanced Frontend Interview Questions

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1. Why does await inside a loop sometimes block UI responsiveness even though JavaScript is single-threaded?
2. Explain a real production issue caused by microtask starvation.
3. What happens internally when a Promise resolves before .then() gets attached?
4. How does V8 optimize hot functions, and what causes de-optimization?
5. Why can delete obj.key negatively impact performance in JS engines?
6. Explain hidden classes and inline caching in V8 with a real example.
7. Why does spreading large arrays (...arr) sometimes crash production apps?
8. What happens in the event loop during heavy React re-renders?
9. Why are for...in loops slower in some cases compared to normal loops?
10. How would you detect memory leaks caused by detached DOM nodes?
11. Explain the complete rendering pipeline from HTML parsing to pixels on screen.
12. What exactly causes layout thrashing?
13. Why does reading offsetHeight after modifying styles trigger reflow?
14. How do browsers batch DOM updates internally?
15. Difference between paint, composite, and layout with real debugging scenarios.
16. What causes GPU layer explosion?
17. Why can transform: translateZ(0) improve performance but also create issues?
18. Explain how Chrome handles compositing layers.
19. What is jank technically inside the browser?
20. Why do animations using top/left lag more than transform animations?
21. How would you debug a page that scrolls smoothly on Mac but lags on Windows?
22. Explain how requestAnimationFrame actually synchronizes with rendering.
23. Why can too many box-shadows reduce FPS?

24. What happens internally when CSS changes on a parent with thousands of children?
25. Explain style recalculation cost in large applications.
26. Explain how React Fiber changed reconciliation internally.
27. What problem does concurrent rendering solve?
28. Why can React render a component multiple times before committing?
29. Difference between render phase and commit phase in React.
30. Explain tearing in React state management.
31. What happens internally when setState is called?
32. Why does React batch updates?
33. Explain how React scheduling priorities work.
34. Why do stale closures happen in hooks?
35. Explain how React detects hook order mismatches.
36. Why can useMemo sometimes worsen performance?
37. What exactly happens when a key changes in React?
38. Why should keys not be indexes in dynamic lists?
39. Explain hydration mismatch in Next.js applications.
40. How does React event delegation work internally?
41. Explain synthetic events vs native events deeply.
42. What happens internally during React hydration?
43. Why can context cause unnecessary re-renders?
44. Explain selector-based state optimization.
45. What problem does useTransition solve internally?
46. How do you identify whether a performance issue is CPU-bound or memory-bound?
47. Explain memory snapshots in Chrome DevTools.
48. What is retained size vs shallow size in heap analysis?
49. How would you debug increasing RAM usage after route changes?
50. Explain garbage collection pauses in frontend apps.

51. Why do closures sometimes accidentally retain huge objects?
52. How can event listeners create memory leaks?
53. Explain how timers can retain memory unexpectedly.
54. How would you debug a React app freezing after 30 minutes?
55. Explain flame charts in the Performance panel.
56. Why is HTTP/2 multiplexing not always faster?
57. Explain how browsers prioritize resource loading.
58. Difference between preload, prefetch, and preconnect.
59. How does code splitting actually work internally?
60. Why can too many chunks hurt performance?
61. Explain tree shaking limitations in modern bundlers.
62. Why do CommonJS modules affect tree shaking?
63. Explain hydration waterfalls in SSR applications.
64. What causes CLS (Cumulative Layout Shift) in production?
65. Explain the browser preload scanner deeply.
66. How would you optimize a dashboard rendering 50 APIs simultaneously?
67. Why can lazy loading hurt LCP sometimes?
68. Explain stale-while-revalidate caching strategy.
69. What happens internally during service worker caching?
70. Explain optimistic UI rollback handling.
71. Why is CSS considered render-blocking?
72. Explain stacking context with a real debugging issue.
73. Why does position: fixed behave differently inside transformed parents?
74. Explain formatting contexts in CSS.
75. What causes massive repaint areas?
76. Difference between flexbox sizing algorithm and grid sizing algorithm.
77. Why do percentage heights fail sometimes?

78. Explain subpixel rendering issues.
79. Why can CSS specificity become a scalability problem?
80. Explain how browsers match CSS selectors internally.
81. Why are descendant selectors expensive?
82. Explain paint containment and layout containment.
83. Explain XSS beyond dangerouslySetInnerHTML.
84. How can frontend source maps become a security risk?
85. Explain prototype pollution in JavaScript.
86. What happens during CORS preflight internally?
87. Explain CSP (Content Security Policy) with real production use cases.
88. How can third-party scripts destroy frontend performance?
89. Explain iframe isolation and sandboxing.
90. What are supply-chain attacks in frontend ecosystems?
91. Design frontend architecture for a scalable eCommerce homepage.
92. How would you build a frontend capable of rendering 100k rows smoothly?
93. Design notification architecture for millions of users.
94. How would you prevent unnecessary re-renders in a trading dashboard?
95. Explain frontend architecture for offline-first applications.
96. How would you design a reusable design system at scale?
97. How would you optimize bundle delivery globally?
98. Design frontend caching strategy for slow networks.
99. How would you architect micro-frontends without performance degradation?
100. Explain frontend monitoring architecture in production.
101. Why does typing sometimes lag in React forms?
102. Explain input latency deeply.
103. Why do large Context providers become performance bottlenecks?
104. Explain browser task queues with real examples.

105. Why does innocent-looking state sometimes trigger huge renders?
106. Explain why virtualization libraries improve performance.
107. What happens when React Suspense pauses rendering?
108. Explain priority inversion in frontend rendering.
109. How do browsers recover from long main-thread blocking tasks?
110. Explain why fast internet still produces slow frontend experiences.
111. How would you debug random FPS drops?
112. Why do some animations become blurry during movement?
113. Explain browser rasterization.
114. What exactly happens when you open Chrome DevTools and performance changes?
115. Why does production behave differently than localhost builds?
116. Explain source map performance impact.
117. How does React DevTools affect rendering?
118. Explain scheduler yielding in concurrent React.
119. Why can debouncing create UX issues?
120. Explain the hidden cost of deep component trees.